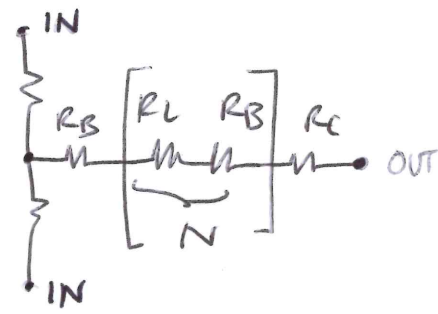
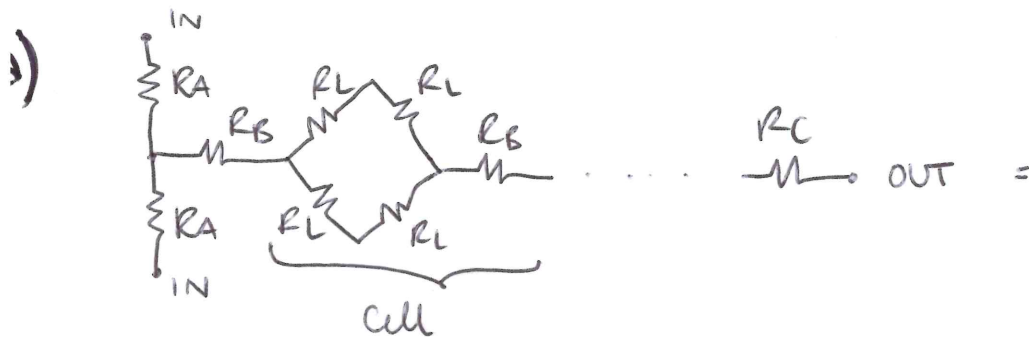


EX. 1

1/4



$\square \begin{matrix} h=10 \\ W=12 \end{matrix} \rightarrow \frac{W}{h} < 1.4 \rightarrow \text{EQUIVALENT RADIUS}$

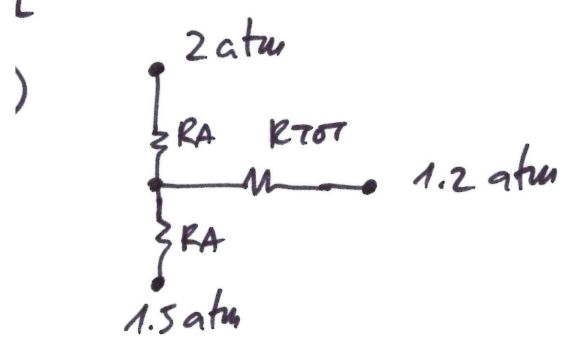
$$r_{eq} = \frac{2 \cdot A}{\text{per}} = \frac{2 \cdot W \cdot h}{2(W+h)}$$

$$= \frac{10 \cdot 12}{(10+12)} = 5.45 \mu m$$

$2 = \gamma \cdot \text{length}$

$$\gamma = \frac{8 \cdot \eta}{\pi (r_{eq})^4} = 2.88 \cdot 10^{18}$$

$2A = \gamma \cdot 3 \mu m = 8.65 \cdot 10^{15} [Pa \cdot s / \mu m^3]$
 $R_B = RA/3 = 2.88 \cdot 10^{15}$
 $R_C = R_B \cdot 2 = 5.77 \cdot 10^{15}$



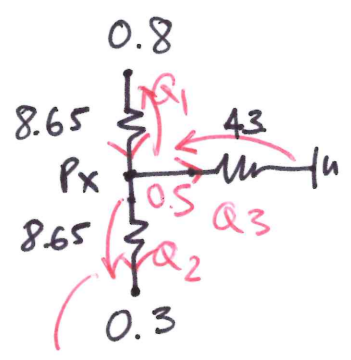
$$R_{TOT} = R_B + R_C + 4(R_L + R_B)$$

$$= 43.25$$

SUPERPOSITION:

$$P_X' = 0.8 \cdot \frac{8.65 / 43.25}{8.65 + 7.2} = 0.36$$

$$P_X'' = 0.3 \cdot \frac{7.2}{8.65 + 7.2} = 0.13$$



$P_X = 0.5$

$$Q_1 = \frac{0.8 - 0.5}{R_A} = 3.5 \cdot 10^{-12} \frac{m^3}{s}$$

$$V_1 = \frac{Q_1}{h \cdot W} = \frac{(29)}{30} \frac{mm}{s}$$

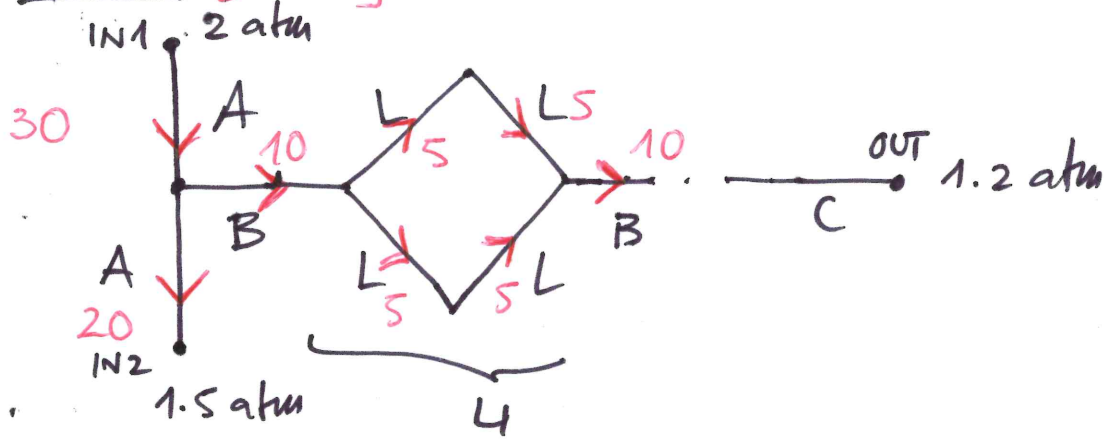
⚠ REVERSE FLOW

$$Q_2 = \frac{0.3 - 0.5}{R_A} =$$

$$V_2 = -19.5 \frac{mm}{s}$$

$$Q_3 = \frac{0.5}{R_{TOT}} = Q_1 - Q_2 = 10 \frac{mm}{s}$$

SPEED: [mm/s]



$$F_{\text{drag}} = 6 \cdot \pi \cdot \eta \cdot r \cdot v =$$

$$v = 1 \text{ cm/s} \quad = 6 \cdot \pi \cdot 10^{-3} \cdot 10 \cdot \text{cm} \cdot 1 \text{ cm/s} = \boxed{18.8 \mu\text{N}}$$

$$F_A - F_g \geq 0$$

Archimede's Gravity

Empty reservoir

$$\frac{4}{3} \pi r^3 \cdot \rho_{\text{H}_2\text{O}} = mg$$

VOLUME FRACTION

$$m = \frac{4}{3} \pi r^3 \cdot \alpha \cdot 3 \cdot \rho_{\text{H}_2\text{O}}$$

$$\frac{4}{3} \pi r^3 \cdot \rho_{\text{H}_2\text{O}} = \frac{4}{3} \pi r^3 \cdot \alpha \cdot 3 \cdot \rho_{\text{H}_2\text{O}}$$

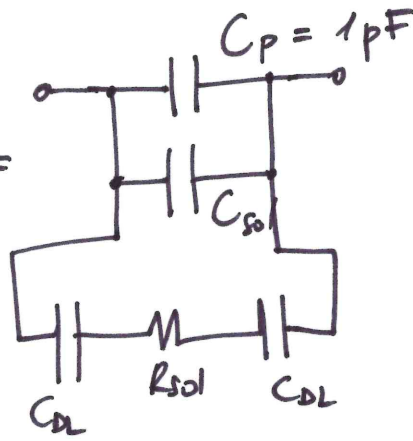
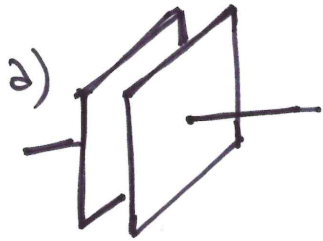
$$\alpha = \frac{1}{3} \Rightarrow \boxed{\text{RESERVOIR VOLUME } \frac{2}{3} \text{ of TOTAL}}$$

2) look at smallest cross-section

$$\frac{10}{12} \Rightarrow \text{to AVOID CLOGGING}$$

FILTER below $10 \mu\text{m}$ with some margin ($\sim 1 \mu\text{m}$)

EX. 2



$$C_{sol} = \epsilon_0 \cdot \epsilon_r \cdot \frac{M \cdot N}{D} = \begin{cases} 22 \text{ pF} \\ 177 \text{ pF} \end{cases}$$

1 - 80
empty full

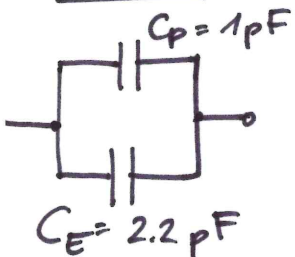
$$C_{dl} = \frac{0.1 \text{ pF}}{\mu\text{m}^2} \cdot \frac{N \cdot M}{10} = 0.10 \text{ mF}$$

$$R_{sol} =$$

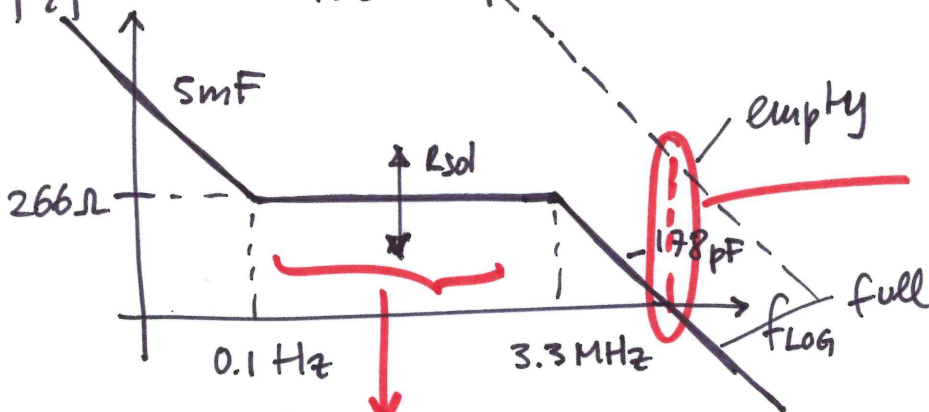
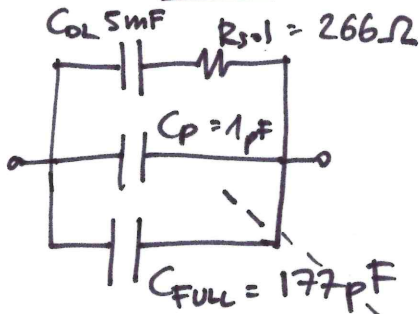
$$C_{dl} \sim \sqrt{\frac{\mu_0}{\sigma}}$$

$$= \frac{100}{1.55/\text{m}} \cdot \frac{D}{M \cdot N} = 266.6 \Omega$$

EMPTY:

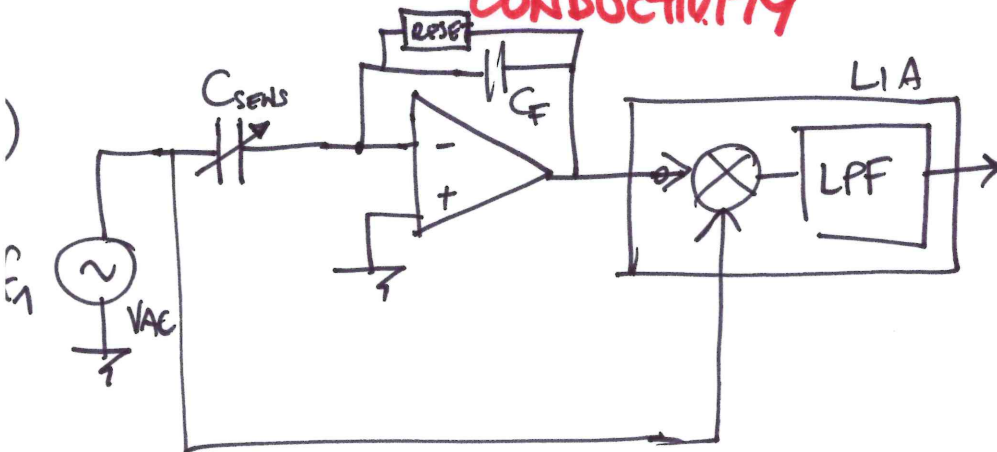


FULL:

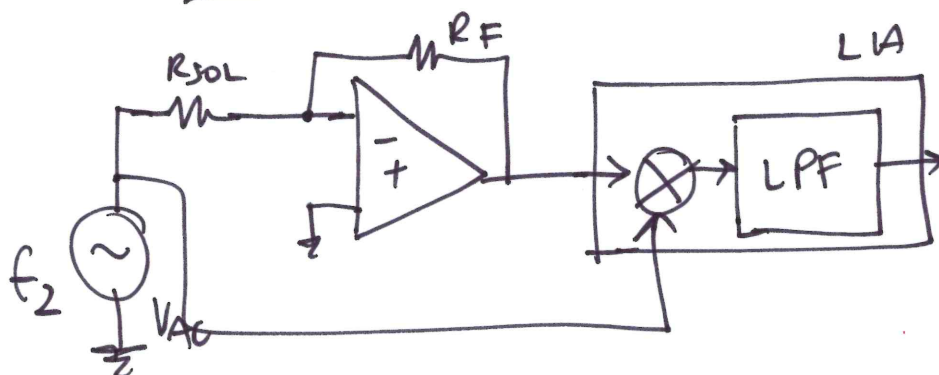


MEASURE CONDUCTIVITY

MEASURE CAPACITANCE
 $f > 10 \cdot 3.3 \text{ MHz}$
 40 MHz
 ↓
 LEVEL



$$f_1 = 40 \text{ MHz}$$



$$f_2 = 100 \text{ Hz}$$

$V_{AC} = 50 \text{ mV}$
 $V_{DD} = +5V$
 $V_{SS} = -5V$
 I choose

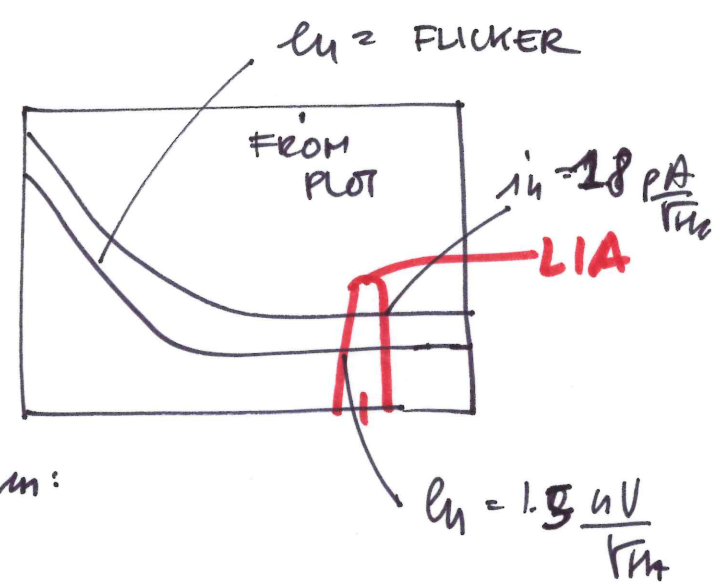
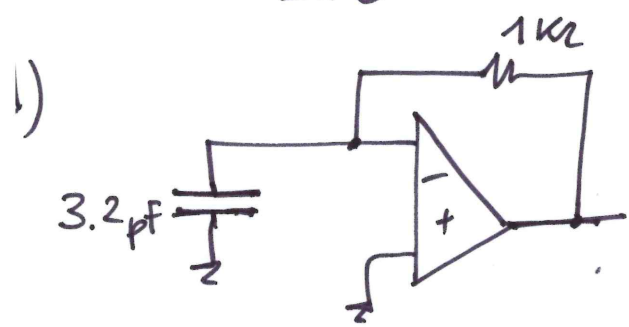
$|V_{out1}| = V_{AC} \cdot \frac{C_{sens}}{C_F} = 5V \Rightarrow$
 max 178 pF

1.78
 $C_F = 1.78 \text{ pF}$
~~TO~~

$|V_{out2}| = V_{AC} \cdot \frac{R_F}{R_{sol}} = 5V \Rightarrow$
 min ASSUME 10 times σ

$R_F = 2.6 \text{ K}\Omega$
 [check BW OK]

1) $BW \sim \frac{1}{2\pi\tau} = 160 \text{ Hz}$



LIA in white region of spectrum:

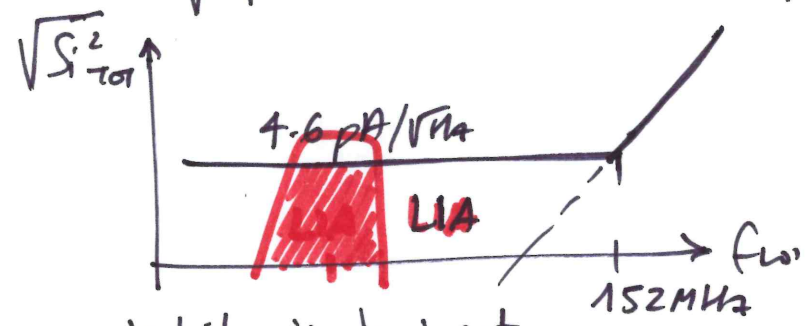
$\sqrt{\frac{4kT}{R_f}} = \sqrt{\frac{4 \cdot \frac{kT}{9} \cdot 9 \cdot \frac{1}{1K\Omega}}{26\mu V}} = 4 \text{ pA}/\sqrt{\text{Hz}}$

$\sqrt{\frac{e_n^2}{R_f^2}} = 1.5 \text{ pA}/\sqrt{\text{Hz}}$

$\sqrt{i_n^2} = 1.8 \text{ pA}/\sqrt{\text{Hz}}$

$\sqrt{(e_n \cdot 2\pi f C_{TOT})^2} =$

TOT white
 $\sqrt{4^2 + 1.5^2 + 1.8^2} = 4.6 \text{ pA}/\sqrt{\text{Hz}}$



White is dominant

$\frac{1.5 \mu V}{\sqrt{Hz}} \cdot 2\pi \cdot f_c \cdot 3.2 \text{ pF} = 4.6 \frac{\text{pA}}{\sqrt{Hz}} \Rightarrow f_c = 152 \text{ MHz} > 40 \text{ MHz}$

2) $PI/C_n \rightarrow$ change of C_{OL}